

Physics-Based Modeling of High-Pressure Gasoline Jet Breakup and Penetration up to 150 MPa

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Abstract

High-pressure liquid jets exhibit complex breakup and mixing behavior governed by turbulence–spray interactions, aerodynamic instabilities, and strong momentum exchange with the surrounding gas. In gasoline compression ignition applications, gasoline is injected at diesel-like pressures, yet its lower density and higher volatility result in spray dynamics that differ substantially from conventional diesel jets. This study presents a computational investigation of high-pressure gasoline sprays at injection pressures of 60, 90, 120, and 150 MPa for multiple injector nozzle geometries using a low-fidelity Reynolds-Averaged Navier-Stokes (RANS) framework. The spray is modeled as a discrete liquid phase with breakup governed by combined Kelvin–Helmholtz and Rayleigh–Taylor instabilities. Simulations are performed in a two-dimensional constant-volume configuration and validated against high-speed shadowgraphy experiments and high-fidelity large eddy simulations (LES). The low-fidelity model accurately reproduces liquid penetration length (LPL) across all cases, with an average deviation of 8.7% relative to experimental measurements. Detailed analysis of pressure, velocity, and turbulent kinetic energy (TKE) fields reveals that at ultra-high injection pressures, LPL is dominated by jet momentum and integral turbulence effects, leaving penetration relatively insensitive to the resolution of small-scale eddies. LES captures finer-scale

spray structures but produces comparable penetration trends to RANS predictions. The results demonstrate that low-fidelity modeling can reliably capture the governing physics of high-pressure gasoline jet penetration while substantially reducing computational cost. The proposed framework provides physically grounded insight into the breakup and dispersion mechanisms of high-pressure liquid sprays and offers a scalable approach for studying momentum-dominated atomizing jets beyond engine-specific applications.

Keywords: Gasoline Compression Ignition; Direct Injection; Spray Characteristics; CFD; Liquid Penetration Length; Spray Angle

Abbreviations

AMR Adaptive Mesh Refinement

ANN Artificial Neural Network

ASOI After Start of Injection

CFD Computational Fluid Dynamics

DNS Direct Numerical Simulations

GDI Gasoline Direct Injection

LES Large Eddy Simulations

LIF Laser-Induced Fluorescence

LPL Liquid Penetration Length

MAE Mean Absolute Error

ML Machine Learning

MPa	Megapascal
PDA	Phase-Doppler Anemometry
PIV	Particle Image Velocimetry
RANS	Reynolds-Averaged Navier-Stokes
ROI	Rate of Injection
TAB	Taylor Analogy Breakup
TKE	Turbulent Kinetic Energy
VOF	Volume of Fluid

1. Introduction

Sprays are a canonical problem in multiphase fluid mechanics, with relevance to a wide range of applications involving atomizing jets, turbulent mixing, and momentum-dominated flow regimes [1–3]. In recent decades, injection pressures in gasoline direct injection (GDI) systems have increased from early values of 5–10 MPa to pressures exceeding 80 MPa, while gasoline compression ignition (GCI) concepts operate at diesel-like pressures approaching 200 MPa [1,2]. At such conditions, gasoline sprays exhibit fundamentally different breakup, penetration, and mixing behavior compared to diesel jets, owing to differences in liquid density, volatility, and interfacial properties [3–5]. These distinctions alter the balance between inertial, aerodynamic, and turbulent forces governing spray evolution, making high-pressure gasoline jets an important case study for momentum-driven liquid jet dynamics.

Increasing injection pressure intensifies surface instabilities and promotes rapid atomization through enhanced Kelvin–Helmholtz and Rayleigh–Taylor mechanisms, leading to smaller characteristic droplet sizes and increased air entrainment [6–10]. At the same time, higher jet momentum accelerates axial

penetration and lateral dispersion, modifying the competition between breakup, turbulent mixing, and wall interaction [11,12]. Variations in injector geometry, including hole number, inlet rounding, and converging or diverging passages, further influence spray morphology by altering near-nozzle flow development and turbulence generation [7,8]. As a result, macroscopic spray characteristics such as liquid penetration length (LPL), spray angle, and plume structure emerge from a complex coupling between nozzle-scale flow, aerodynamic instabilities, and ambient gas dynamics.

Experimental investigations have provided valuable insight into spray structure and breakup processes using optical diagnostics such as laser-induced fluorescence (LIF), particle image velocimetry (PIV), high-speed imaging, and phase-Doppler anemometry (PDA) [14–17]. However, experimental characterization of high-pressure sprays remains challenging due to optical access limitations, transient internal nozzle flow, and the highly unsteady nature of multiphase jet breakup, particularly under flash-boiling and ultra-high-pressure conditions [18–21]. These constraints complicate the development of generalized empirical correlations, as spray behavior inferred from one configuration may not readily extrapolate to other geometries or operating regimes [20].

Numerical studies using high-fidelity approaches, such as direct numerical simulations (DNS) and large eddy simulations (LES), have significantly advanced the understanding of spray physics by resolving internal nozzle flow, primary breakup, and turbulence–spray interaction. For example, Li et al. employed DNS with a volume of fluid (VOF) method to capture primary and secondary atomization, demonstrating the influence of internal turbulence and cavitation on jet disintegration [31]. Leng et al. investigated modified injector geometries and reported measurable reductions in characteristic droplet size with increasing injection pressure, highlighting the sensitivity of breakup dynamics to near-nozzle flow structures [32]. Using LES, Hu et al. showed that subtle changes in orifice inlet geometry can substantially alter spray plume development and turbulence generation [33]. While these studies provide detailed physical insight, their computational cost limits their applicability for systematic exploration across wide ranges of injection pressure and nozzle design [2,36].

In parallel, data-driven approaches, including machine learning (ML) models, have been applied to predict macroscopic spray metrics such as penetration length and cone angle with low computational cost [37–42]. For example, Khan et al. deduced that using a tree-ensemble ML model can result in accurate predictions of spray penetration and angle with mean absolute error (MAE) of 0.651 [43]. Richards and Emekwuru developed an artificial neural network (ANN) model capable of predicting vapor tip penetration across multiple fuels with an R^2 of 0.999, demonstrating its potential for multi-fuel applications [37]. While these methods can achieve high accuracy, their predictive robustness is often limited by data scarcity and the absence of explicit physical constraints, particularly when extrapolated to ultra-high-pressure regimes [44]. Classical penetration correlations, including those proposed by Wakuri, Dent, and Hiroyasu–Arai, provide valuable physical insight but were developed for lower injection pressures and do not fully capture the physics of modern high-momentum sprays [45–47].

Motivated by these limitations, the present work employs a low-fidelity Reynolds-Averaged Navier-Stokes (RANS) framework to investigate the penetration and breakup behavior of high-pressure gasoline jets over a wide range of injection pressures and nozzle geometries. By validating the model against high-speed experimental measurements from a constant-volume configuration [48], the study demonstrates that macroscopic spray penetration is governed primarily by jet momentum and integral turbulence effects, and is relatively insensitive to small-scale eddy resolution. This approach enables efficient, physics-consistent exploration of high-pressure spray regimes and provides a robust foundation for the development of generalized penetration correlations applicable to momentum-dominated liquid jets.

2. Methodology

The methodology of the current study includes the development of a computational fluid dynamics (CFD) code for high-pressure gasoline sprays using CONVERGE CFD software, which is validated by comparison with experimental data. The injector design, chamber, and operating conditions considered in the study are depicted from the experiments conducted by Medina et al. [4] in a constant-volume chamber filled with nitrogen maintained at 298 K. As shown in Figure 1, diffuse backlight shadow imaging was employed to

observe the spray development, using a high-speed camera positioned perpendicular to the injector axis. Reference-grade gasoline blended with 350 ppm of lubricant was used to comply with the requirements of the Bosch CP3 pump that was used. Experimental studies reported an uncertainty of ± 0.2 bar in the chamber pressure and ± 10 bar in the injection pressure, with a fixed chamber temperature of 298 K and an associated uncertainty of ± 3 K. Further details of the experimental setup can be found in Medina et al. [4]. The following subsection presents the design and operating conditions, the governing equations implemented in the CFD code, and a discussion of the geometry, mesh settings, validation, and results.

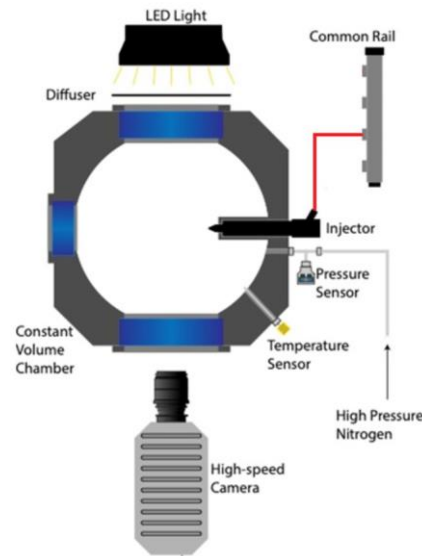


Figure 1. High speed imaging setup for capturing spray development in a constant chamber vessel filled with nitrogen maintained at different ambient pressure and fixed 298K temperature [4]

2.1. Design and Operating Conditions

Three different injector geometries, designated X, Y, and Z, are considered in the current study [4]. The injectors varied in terms of the number of nozzles, converging/diverging passages, inlet rounding, and outlet diameter. Injector X has one hole while injector Y and Z have two holes. As shown in Figure 2, the holes

for injector Y and Z are located 15° above and below the plane and separated by 30°. For clarity, Y1 and Y2 refer to hole 1 and hole 2 of injector Y, respectively. Similarly, Z1 and Z2 correspond to hole 1 and hole 2 of injector Z, respectively. An overview of the injector geometries used in the experiments is summarized in Table 1. The nozzle outlet diameter is normalized by 190 μm, which is the maximum outlet diameter across all injectors. The conicity is determined by the K factor based on d_i (inlet diameter) and d_o (outlet diameter) as defined in Eq. 1.

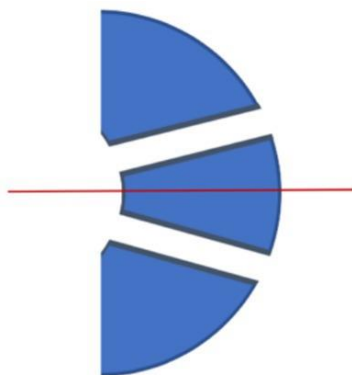


Figure 2. Schematic of the dual-hole orifice for injector Y and Z where the holes are separated by 30° and are located 15° each above and below the plane [4]

$$K \text{ factor} = \frac{(d_i - d_o)}{10} \quad \text{Eq. 1}$$

Table 1. Summary of the dimensions and types of injectors X, Y and, Z used in experiments. The schematic of the nozzle orifice is adapted from Medina et al. [4]

Injector	Number of holes	Hole type	Outlet diameter (normalized)	$K - factor$	Schematic
X	1	Converging	0.578	1.5	

Y	2	Y1-Straight	0.578	0	
		Y2-Straight	1	0	
Z	2	Z1-Diverging	0.789	-1.5	
		Z2-Converging	0.789	3.5	

The domain is filled with nitrogen as the gas phase at 0.1 MPa, and the chamber temperature is maintained uniformly at 298 K. Iso-octane (iC_8H_{18}) is injected into the domain as parcels containing drops. Since the spray is non-reacting, the chemical mechanism requires only four elements and five species. The elements include C, H, N and O while the species are iC_8H_{18} , N_2 , O_2 , H_2O , and CO_2 . Fuel injection is carried out at pressures of 60, 90, 120, and 150 MPa. Since the injection duration is 1 ms, a minimum time step of 1×10^{-11} ms is set to accurately capture the transient behavior of the spray. The operating conditions are summarized in Table 2.

Table 2. Summary of operating conditons used across all the injectors X, Y, and Z.

Operating Conditions	Value
$t_{\text{injection}}$ [ms]	1
P_{ambient} [MPa]	0.1
$P_{\text{injection}}$ [MPa]	60, 90, 120, 150
Ambient temperature [K]	298

2.2. Governing Equations and Solver Settings

The fuel is injected as drop parcels at the injector location based on the Rate of Injection (ROI) profile. These parcels containing drops, identical in size and properties, are used to represent the entire spray. For

each drop, the diameter is assumed to be equal to the nozzle outlet diameter. The contraction effect of the nozzle is accounted for using the discharge coefficient (C_d), which is directly obtained from experimental measurements. The velocity coefficient (C_v) is then derived from the injection pressure within the CFD code. The contraction coefficient (C_a) is defined by Eq. 2.

$$C_a = \frac{C_d}{C_v} \quad \text{Eq. 2}$$

The discharge coefficient as a function of Reynolds number for injectors Y and Z is shown in Figure 3. Since injector Z has a larger inlet rounding than injector Y, its discharge coefficient is significantly higher. However, the C_d remains similar for a range of Reynolds number.

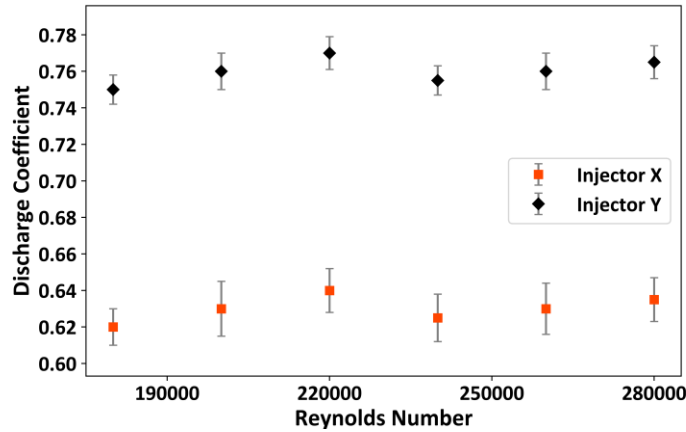


Figure 3. Discharge coefficient as a function of Reynolds number for dual hole injectors Y and Z derived from experiments. The error bars represent the standard deviation from the mean across 20 experimental cycles [4]

The ROI profile obtained from experiments for injector Y at 150 MPa injection pressure and 0.1 MPa chamber pressure is shown in Figure 4. The shaded area represents the total mass injected from both the holes during the period of injection starting from the needle lift.

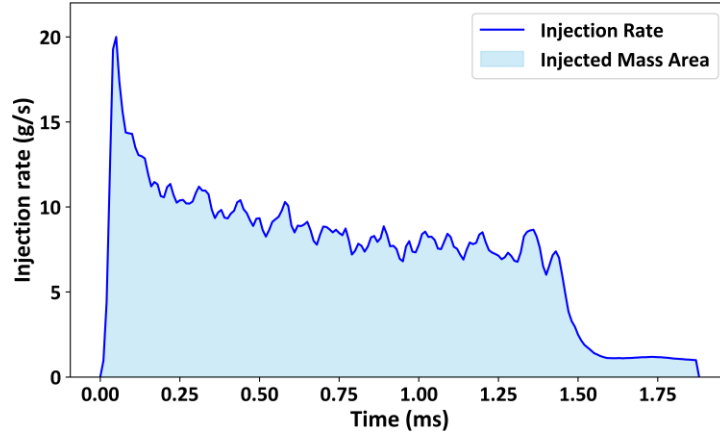


Figure 4. ROI profile for injector Y with two nozzles. Fuel is injected at 150 MPa, and the back pressure is maintained at 0.1 MPa. The shaded area corresponds to the total injected mass[4]

The velocity of the droplet having a density ρ_l and volume V_d is given by the Eq. 3.

$$\rho_l V_d \frac{dv_i}{dt} = F_{d,i} \quad \text{Eq. 3}$$

v_i is the velocity of the drop and $F_{d,i}$ gives the total drag and gravitational force experience by the drop (Eq. 4).

$$F_{d,i} = F_{drag,i} + F_{g,i} \quad \text{Eq. 4}$$

The drag coefficient on the drop varies continuously since the drop undergoes continuous distortion throughout the course of injection. It is calculated based on the Taylor Analogy Breakup (TAB) model.

$$C_D = C_{D,sphere}(1+2.632y) \quad \text{Eq. 5}$$

In Eq. 5, C_D is the drag on the drop, $C_{D,sphere}$ is the drag on a spherical drop and, y is the distortion parameter calculated based on the TAB model. $C_{D,sphere}$ can be estimated by the following equation

$$C_{D,sphere} = \begin{cases} 0.424, & Re \geq 1000 \\ \frac{24}{Re} \left(1 + \frac{1}{6} Re^{2/3}\right), & Re \leq 1000 \end{cases} \quad \text{Eq. 6}$$

The breakup of the jet due to instabilities is modeled using the Kelvin-Helmholtz (KH) and Rayleigh-Taylor (RT) analogies. The instability wavelengths in the KH and RT models are based on the formulations proposed by Reitz and Xin et al. respectively [5,6]. The constants associated with the KH-RT models are kept as default values. The breakup model involves the transformation of a liquid jet into liquid lumps due to aerodynamic instabilities, followed by their subsequent conversion into droplets induced by air entrainment. In addition to breakup, the collision of the droplets is modeled using the no time counter method of Schmidt and Rutland [7].

The effect of turbulence on the droplet is modeled by adding the fluctuating velocity (u'_i) to the gas velocity (u_i). Since RANS is used for turbulence modeling, the depletion of turbulent kinetic energy (TKE) due to the dispersion of spray droplets is accounted for in the code by incorporating source terms (S_s). These source terms are summed over all parcels containing N_p droplets within the cell volume V .

$$S_s = - \frac{\sum_p N_p (F'_{drag,i} u'_i)_p}{V} \quad \text{Eq. 7}$$

The fluctuation in the drop drag force is given by Eq. 8.

$$F'_{drag,i} = \frac{F_{drag,i}}{(u_i + u'_i - v_i)} u'_i \quad \text{Eq. 8}$$

The liquid fuel is injected into an ambient atmosphere with a uniform temperature of 298 K. The resulting spray is non-reacting, and the LPL is calculated using the CONVERGE code. Spray-wall interaction is modeled using the O'Rourke method, which considers the Weber number, film thickness, and viscosity [8]. The summary of models associated with spray and simulation time parameters has been summarized in Table 3. To further analyze the effect of turbulence in the flow regime, LES was performed and compared

with RANS results for nozzles Z1 and Z2 at different after start of injection (ASOI) times under the same operating conditions.

Table 3. Sumamry of solver settings in CONVERGE

Time step/ total time [ms]	$1 \times 10^{-11} / 1$
Scheme	PISO
Turbulence model	RANS Standard $k - \epsilon$
Spray model	Discrete Phase
Breakup model	KH-RT
Drop drag	Dynamic drag
Collison	NTC
Turbulent dispersion	O'Rourke
Wall-film interaction	O'Rourke

2.3. Geometry and Mesh

A two-dimensional domain measuring $100 \text{ mm} \times 100 \text{ mm}$ is used for the spray simulation, as shown in Figure 5. The domain is automatically meshed by the CFD code during runtime with an optimal base grid size of 2 mm. Fixed embedding is applied at the injector location to resolve the near-nozzle flow field. In addition, Adaptive Mesh Refinement (AMR) is employed for velocity and species fields, with a maximum embedding level of 3, to optimize computational efficiency. The mass fraction contour of iso-octane 0.6 ms ASOI is shown in Figure 5.

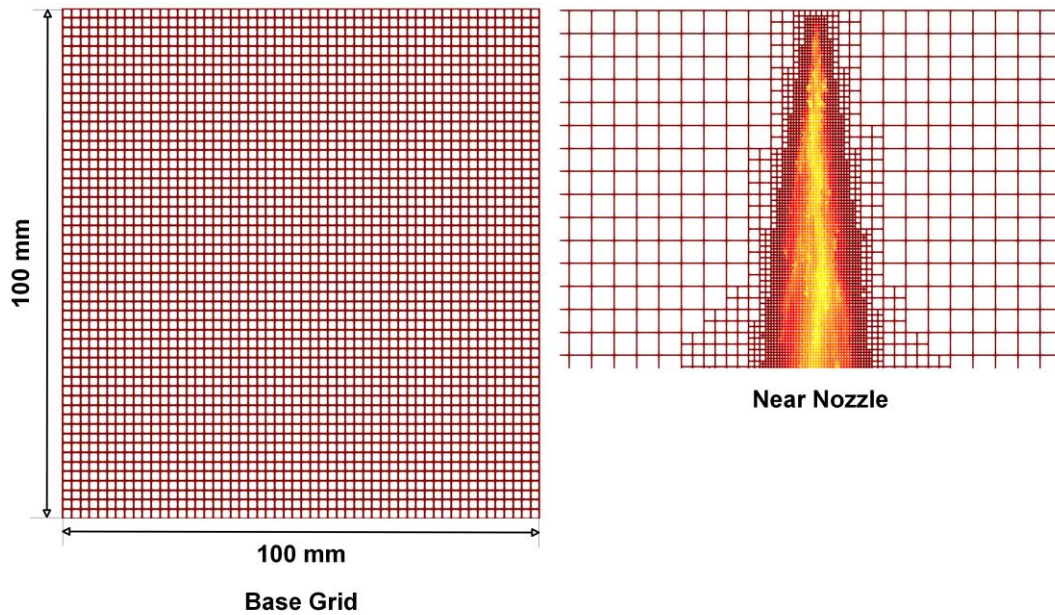


Figure 5. Computational domain measuring 100×100 mm with a base grid size of 2 mm (left), and near-nozzle grid at 0.6 ms ASOI showing a minimum grid size of 0.25 mm (right).

3. Results and Discussion

The results of the simulation are validated against the LPL obtained from experimental data for the simulated cases. The following sections present the validated cases, followed by a detailed discussion on the effects of operating and design conditions on spray characteristics. Parameters such as pressure, velocity, and TKE fields are analyzed to provide insights into the differences between high- and low-pressure sprays.

3.1. Validation

The computational spray model for injector Y1 is compared against experimental data at 0.2 ms ASOI as shown in Figure 6. The ambient pressure is maintained at 0.1 MPa, while fuel is injected at 150 MPa. The mass fraction contour of iso-octane obtained from the model is qualitatively comparable to the spray from experiments. Quantitative comparison on the other hand shows that the LPL closely matches the

experimental data within the experimental tolerances throughout the injection event as shown in Figure 7. Due to optical limitations in the experimental setup, the maximum measurable penetration is limited to 35 mm. This shows that more insights can be gained using the simulation results, overcoming the optical access limitation.

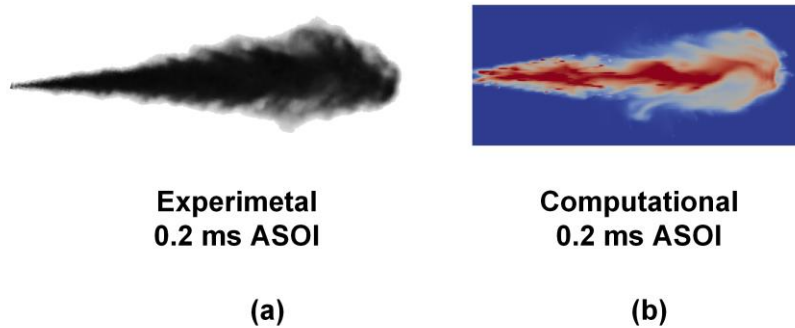


Figure 6. (a) Experimental spray image and (b) spray mass fraction contour at 0.2 ms ASOI for injector Y1. Fuel is injected at 150 MPa, and the back pressure is maintained at 0.1 MPa.

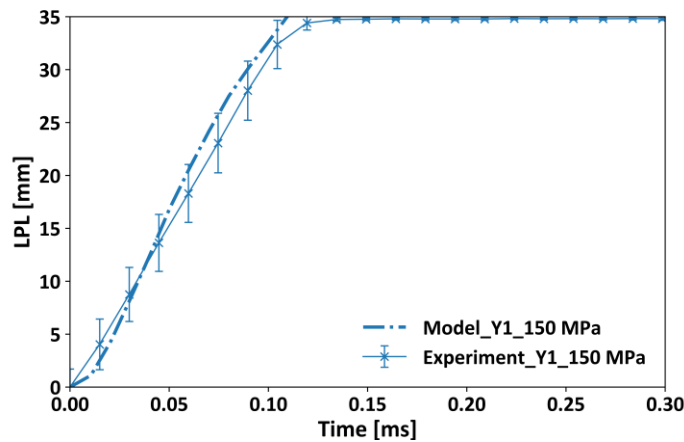


Figure 7. Comparison of LPL between experimental data and the model for Y1 nozzle at 150 MPa injection pressure. The error bars represent the standard deviation from the mean across 20 experimental cycles.

The spray angle has been validated across the two nozzles Z1 and Z2 for injection pressures of 60 MPa. The spray angle is measured at one third of the maximum penetration as done in the experimental study

and is shown in Figure 8. The differences between the computational and experimental spray angle are reported in Table 4 and this difference can arise due to near-nozzle flow and internal geometry effects that are not captured completely by the current turbulence model.

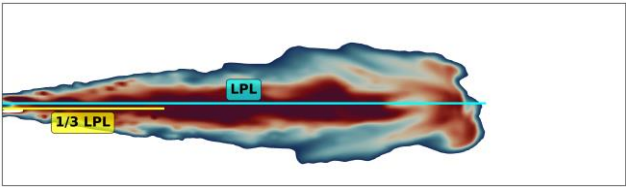
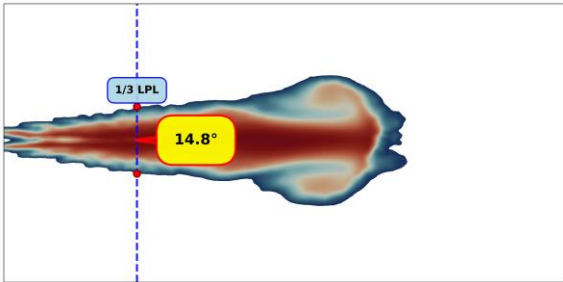
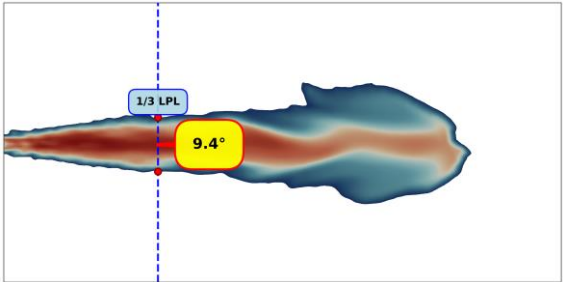


Figure 8. Spray angle measurement from spray mass fraction contour for injector nozzle Y2 at 150 MPa injection pressure

Table 4. Spray angle measurements from computational spray contours for nozzles Z1 and Z2 at injection pressure of 60 MPa.

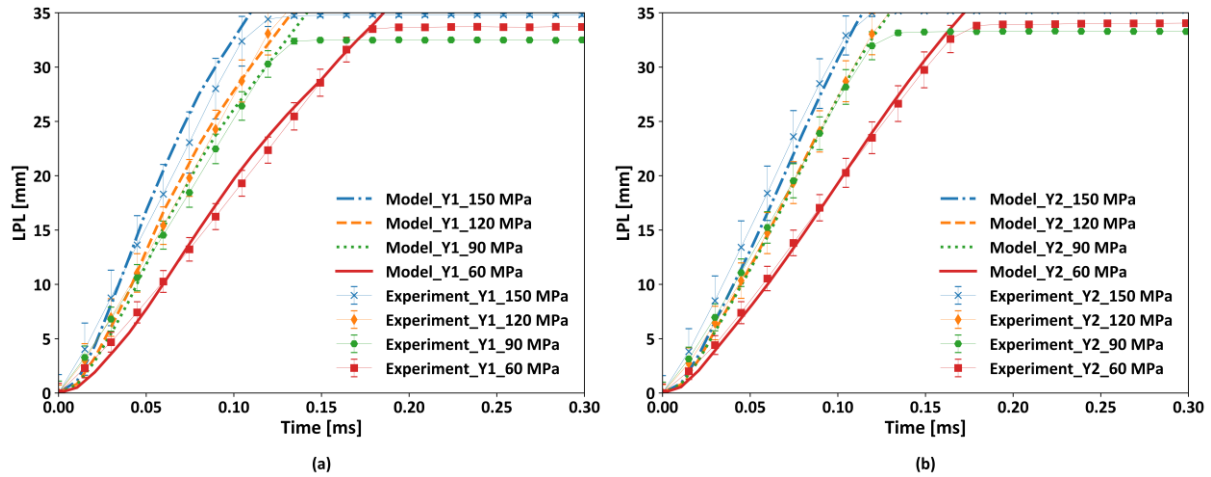
Injector	Computational Spray Angle	Experimental Spray Angle
Z1, 60 MPa injection pressure, 0.3 ms ASOI		16.7
Z2, 60 MPa injection pressure, 0.3 ms ASOI		12.1

3.2. Effect of Injection Pressure

The LPL as a function of injection pulse duration for injection pressures for different injector geometries and injection pressures is presented in Figure 9 (a)-(d). The results show that higher injection pressures lead

to faster penetration at all times due to increased spray momentum. Cases with injection pressures of 90, 120, and 150 MPa demonstrated good agreement with experimental results across all nozzles.

At the lower injection pressure of 60 MPa, the straight nozzles Y1 and Y2 showed excellent agreement with experimental data and remained well within the standard deviation (see Figure 9 (a) and Figure 9 (b)). However, for the Z1 nozzle slight underprediction near the nozzle and slight overprediction downstream were observed as shown in Figure 9 (c) and Figure 9 (d). This behavior is owed to flow expansion at the nozzle exit leading to reduced turbulence intensity. In addition, delayed primary breakup results in a longer coherent liquid core causing higher downstream LPL compared to the experimental measurements. For the Z2 nozzle, the downstream overprediction can be attributed to localized acceleration of the jet due to the converging nozzle geometry.



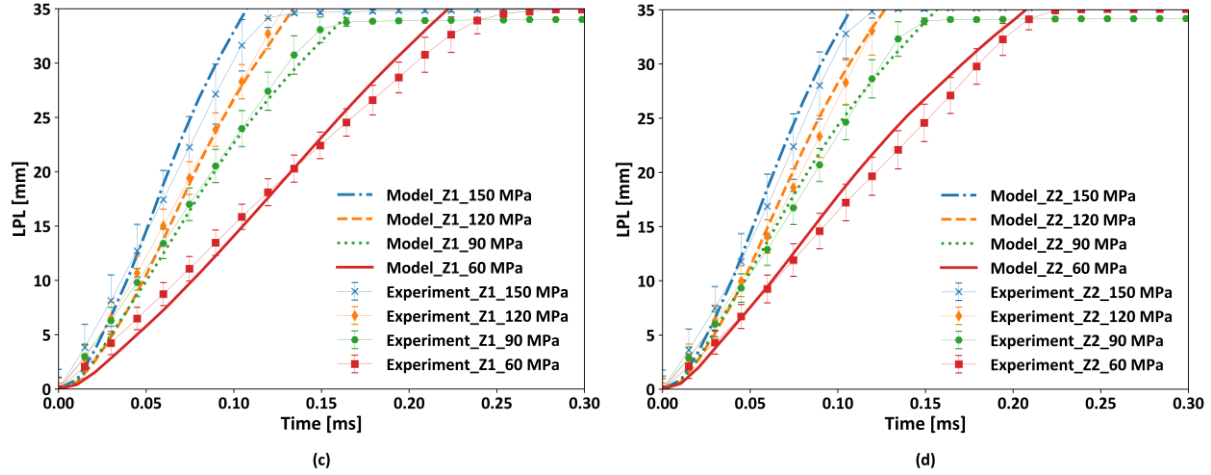


Figure 9. Comparison of LPL between experimental data and model for nozzles (a) Y1 (b) Y2, (c) Z1, and (d) Z2 at injection pressures 60, 90, 120, and 150 MPa. The error bars represent the standard deviation from the mean across 20 experimental cycles.

When the spray is injected at different injection pressures, it exhibits distinct expansion behavior within the chamber. The variation in pressure oscillations can be observed in Figure 10, where for an injection pressure of 150 MPa into a 0.1 MPa chamber, larger pressure gradients are evident near the nozzle. The magnitude of these pressure variations indicates that the jet velocity is sufficiently high to induce compressibility effects in the surrounding gas. In contrast, for the 60 MPa case, the pressure remains close to ambient which suggests slower mixing and weaker dynamic interaction. The pressure contours show that at 150 MPa, strong pressure gradients are developed near the nozzle which is followed by significant compression regions downstream (see Figure 11). These are caused by the high-momentum jet decelerating against the ambient gas. This leads to pronounced turbulent effects and vigorous interaction between the spray and the surrounding medium. However, the expansion intensity is lower for 60 MPa jet and the pressure field appears more uniform and smoothly diffused. The spray regime for 150 MPa pressure results in better mixing and promotes efficient and stable combustion compared to 60 MPa pressure.

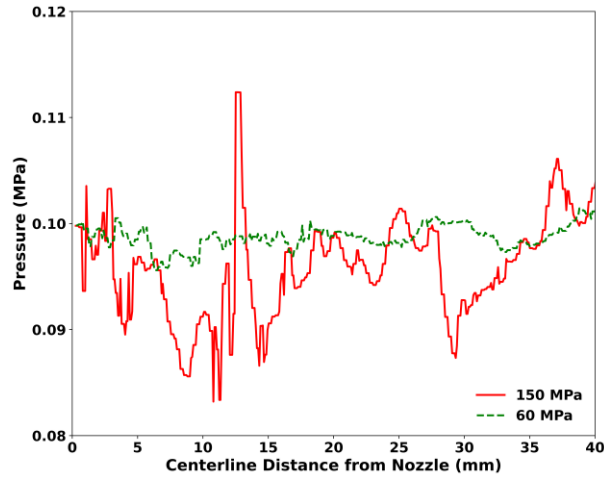


Figure 10. Pressure distribution along the centerline from the nozzle (Z2) for injection pressures of 60 MPa and 150 MPa.

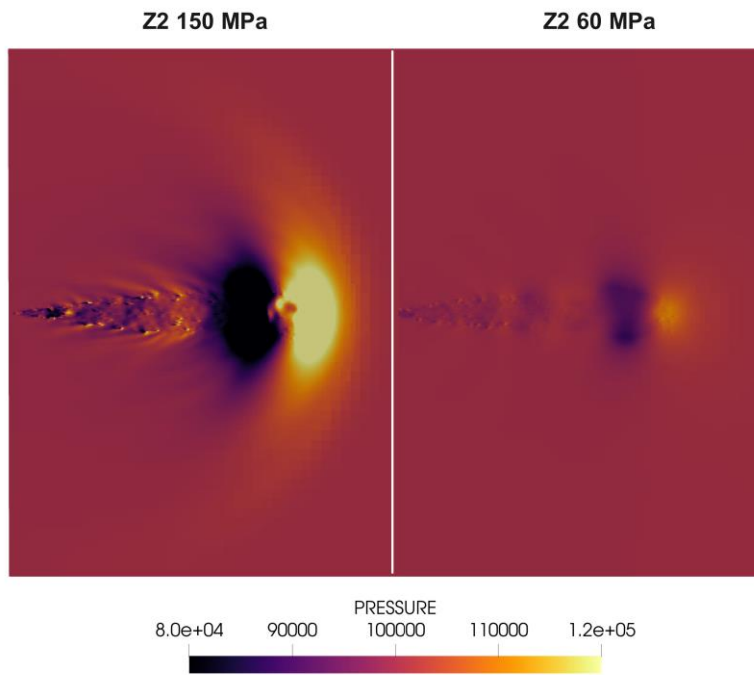


Figure 11. Pressure distribution contours along the centerline from the nozzle (Z2) for injection pressures of 60 MPa and 150 MPa. The color scale is normalized to the same pressure range for direct comparison.

The penetration rate profile at high and low injection pressures of 150 MPa and 60 MPa respectively is illustrated in Figure 12. The penetration rate begins at zero since penetration is zero at the starting time and swiftly attains a maximum value. Increasing injection pressure from 150 MPa to 60 MPa significantly enhances the penetration rate due to higher jet momentum. Consequently, peak penetration occurring earlier for high injection pressure indicates faster spray breakup. At early injection times, Y1 and Y2 exhibit similar penetration rates and later diverge due to entrainment effects. Since Y1 has a smaller outlet diameter compared to Y2, the penetration rate is faster at injection pressure of 150 MPa.

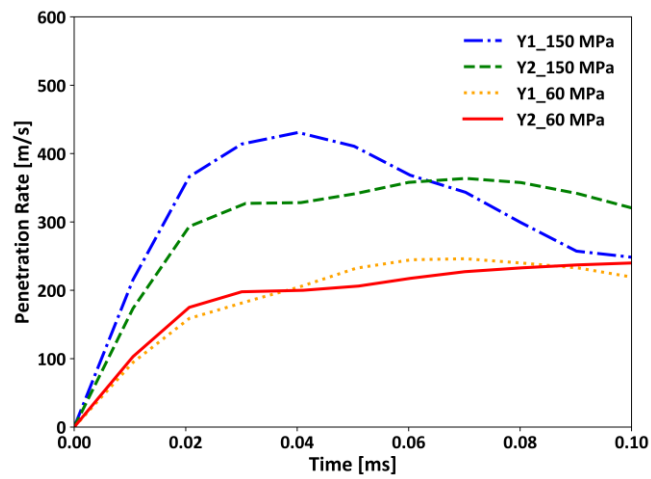


Figure 12. Penetration rate profile derived from penetration for injection pressures of 60 MPa and 150 MPa for nozzles Y1 and Y2.

The break-up time, defined as the duration corresponding to the maximum penetration rate, is depicted in Figure 13 for all nozzles. The decreasing breakup time with increasing injection pressure shows that instability in the spray leads to faster disintegration and consequently early breakup. This observation can also be deduced from the penetration rate curves. In case of nozzle Y1, shortest breakup time is observed across all injection Pressures whereas Y2 nozzle maintains the coherent structure for slightly longer time than Y1 nozzle except at 150 MPa injection pressure. This behavior can be attributed to a larger diameter hole of Y2 nozzle which preserves the axial momentum and resists the jet breakup. However, at 50 MPa injection pressure, the role of geometry diminishes in dictating the breakup.

For the case of Z1 nozzle that is diverging in shape, the onset of breakup is late at injection pressure of 60 MPa when compared across all nozzles. This indicates that the core of the spray is intact for a longer time before instabilities arise. In contrast, Z2 nozzle shows a much earlier breakup at 60 MPa due to converging shape that leads to faster acceleration and destabilization. With increasing injection pressures, the differences in breakup time reduces which again indicates that geometry effects are less dominant than operating conditions.

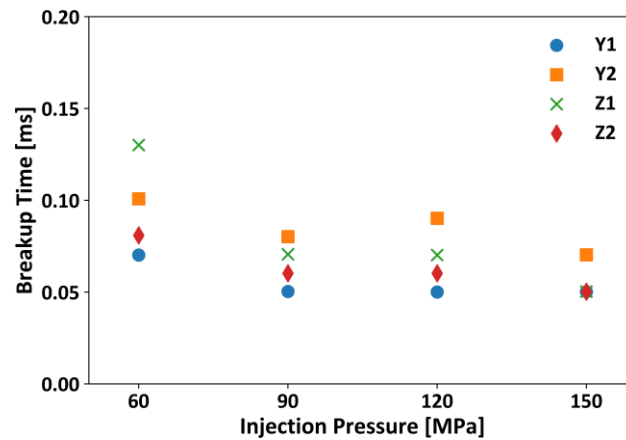


Figure 13. Breakup times across different injection pressures for all nozzles.

3.3. Effect of Nozzle Geometry

For the straight nozzles, Y2 has a larger outlet diameter than Y1, resulting in a faster LPL across the 150 MPa injection pressure. At the lower injection pressure of 60 MPa, the LPLs of Y1 and Y2 were similar during the early stages of injection, indicating that the effect of geometry was relatively insignificant compared to the influence of injection momentum (see Figure 14 (b)). At the higher injection pressure of 150 MPa, the converging nozzle (Z2) and diverging nozzle (Z1) exhibited similar LPLs, as shown in Figure 14 (b). For times beyond 0.15 ms ASOI, this trend remains consistent, as illustrated in Figure 15. This suggests that the high jet momentum at higher pressures diminishes the influence of nozzle geometry on spray penetration during the early phase of injection. However, at the lower injection pressure of 60 MPa,

nozzle Z2 showed a comparatively longer LPL than nozzle Z1. This can be attributed to the diverging shape of Z1, which promotes spray dispersion and reduces axial momentum.

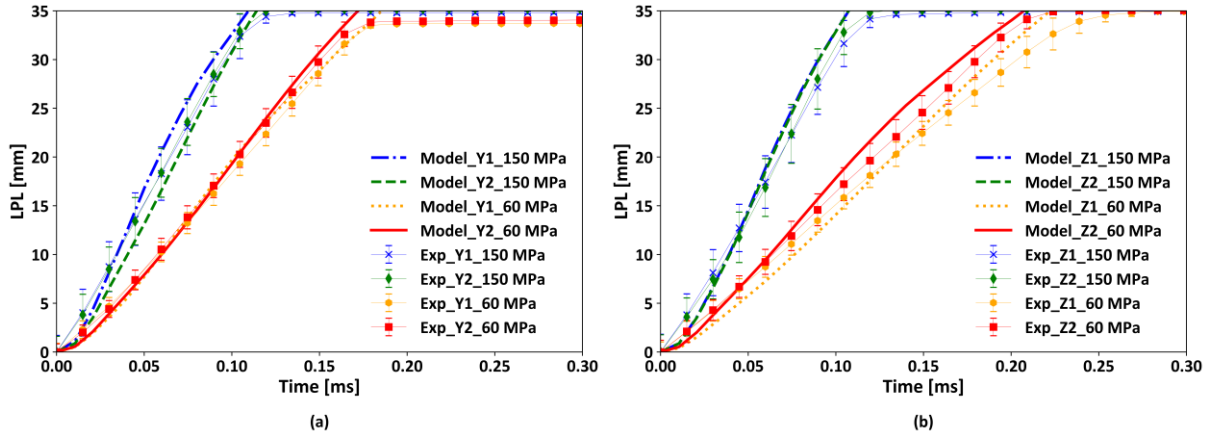


Figure 14. LPL comparison of experiment and model for nozzles (a) Y1/Y2 and (b) Z1/Z2 at 60 MPa and 150 MPa injection pressure. The error bars represent the standard deviation from the mean across 20 experimental cycles.

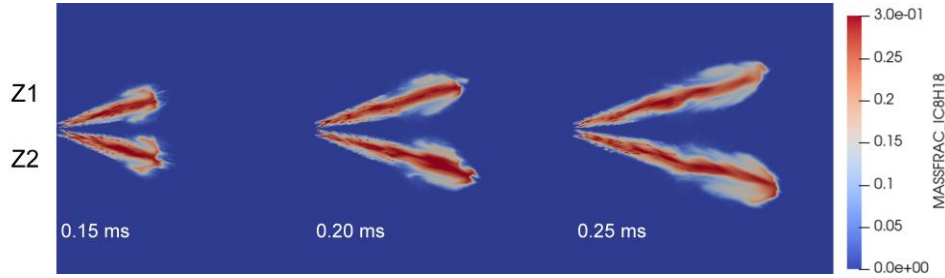


Figure 15. Comparison of spray development for nozzles Z1 and Z2 at different ASOI. The contours represent the mass fraction of fuel injected into a nitrogen-filled ambient.

3.4. Effect of Turbulence Models and Turbulence Intensity

To analyze the effect of the turbulence model on the behavior of injected liquid parcels, an LES model was employed and compared with the RANS approach. The LES model resolves the flow field and captures fine-scale turbulent structures, particularly at later times of 0.25 ms and beyond. However, the RANS model

produces symmetric structures and fails to resolve smaller eddies. As a result, LES provides a more realistic prediction of jet breakup (see Figure 16). Despite differences in the overall spray structure, the spray penetration remains approximately the same for the given injection duration. However, notable differences arise when microscopic characteristics are required, which is beyond the scope of this study.

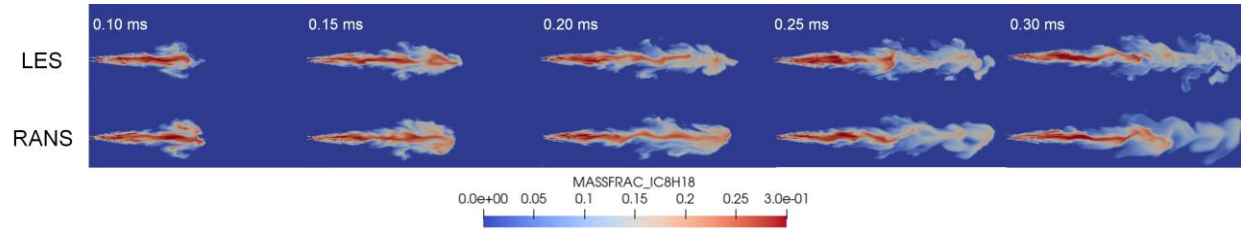
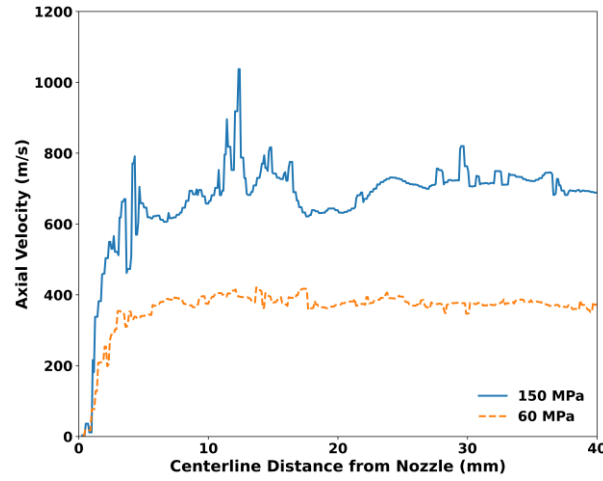


Figure 16. Comparison of spray development for nozzles Z1 and Z2 using LES and RANS models at different ASOI. The contours represent the mass fraction of fuel injected into a nitrogen-filled ambient.

Since the RANS model provides an accurate representation of the LPL under various conditions, a comparison of the turbulence effects was carried out to understand the spray breakup at different injection pressures. At a high injection pressure of 150 MPa, the spray velocity is significantly higher with an average velocity of 680 m/s, leading to stronger velocity fluctuations. This behavior can be observed in Figure 17, where the centerline spray velocity at 150 MPa is greater compared to that at 60 MPa, and the fluctuations are more pronounced due to enhanced turbulence. Consequently, the TKE is higher and concentrated near the nozzle, resulting in core destabilization. Owing to the larger pressure difference between the fuel and the ambient environment, turbulence also persists further downstream, promoting faster breakup and the formation of smaller droplets.

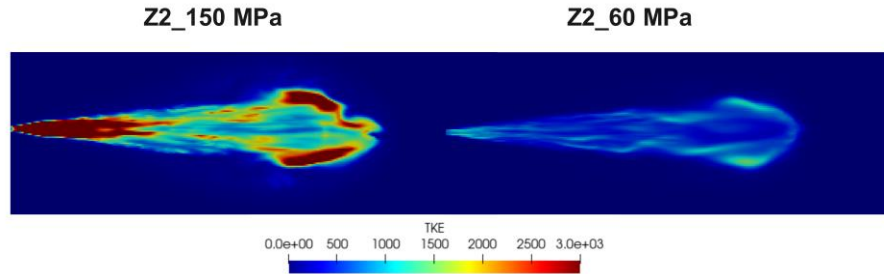
As illustrated in Figure 18, the TKE contours for the high injection pressure (150 MPa) and the low injection pressure (60 MPa) are compared for the same amount of injected mass. At the lower injection pressure, the breakup process is slower because the TKE remains localized near the nozzle tip. The jet is narrower due to reduced velocity fluctuations, leading to slower mixing with the surrounding gas.

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333

334 Figure 17. Axial velocity magnitude measured from the nozzle centerline for injection pressures of 150 MPa and 60
335 MPa.



336

337 Figure 18. Comparison of TKE contours for sprays at injection pressures of 150 MPa and 60 MPa, for the same
338 amount of injected mass. The color scale is normalized to the same TKE range for both cases.

339 4. Conclusions

340 A computational framework based on low-fidelity Reynolds-Averaged Navier-Stokes (RANS) modeling
341 was developed and validated against high-pressure gasoline spray experiments spanning multiple injector
342 geometries and injection pressures from 60 to 150 MPa. The simulations accurately reproduced liquid
343 penetration length across all tested conditions, demonstrating that RANS-based approaches can reliably
344 capture macroscopic spray penetration behavior in ultra-high-pressure injection regimes. Analysis of

operating and geometric effects shows that at elevated injection pressures, spray penetration is governed primarily by jet momentum and integral turbulence effects, resulting in reduced sensitivity to nozzle geometry. At lower injection pressures, geometric features such as converging and diverging passages exert a more pronounced influence on penetration behavior through their impact on near-nozzle flow development and turbulence generation. Compared with low injection pressure sprays, higher injection pressure sprays exhibit relatively higher jet velocities, which promote faster jet breakup. Under non-evaporating conditions and at sufficiently high injection pressures, gasoline sprays can behave in a diesel-like manner, where the jet is primarily momentum-driven. Therefore, diesel spray penetration correlations can be reasonably applied to gasoline sprays under these conditions. Comparisons with large eddy simulations (LES) reveal that while LES resolves finer-scale turbulent structures and more detailed spray morphology, the predicted liquid penetration length remains largely unchanged relative to RANS predictions, indicating that penetration is insensitive to the resolution of small-scale eddies in momentum-dominated regimes. These findings clarify the physical limits of modeling fidelity required for predicting macroscopic spray penetration and highlight the dominant role of jet-scale dynamics over detailed turbulence structure in high-pressure liquid jets. The validated low-fidelity framework enables efficient, physics-consistent exploration of spray behavior across wide parameter spaces that are impractical for high-fidelity simulations or experiments alone. As such, the present work provides a robust foundation for developing generalized penetration correlations and reduced-order descriptions applicable to momentum-dominated multiphase jets, and contributes to a broader understanding of high-pressure liquid spray physics.

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Author Declarations

370 The authors have no conflicts to disclose.

371 **Author Contributions**

372 **Fahad Alzahrani:** Writing – original draft, Conceptualization, Methodology, Investigation, Formal
373 Analysis, Supervision, Funding acquisition. **Sadique Khan:** Writing – original draft, Validation, Formal
374 Analysis, Visualization, Investigation. **James W.G. Turner:** Writing – review & editing, Supervision,
375 Funding acquisition.

376 **Data Availability**

377 The data that support the findings of this study are available from the corresponding author upon reasonable
378 request.

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